

saliyah at StanceEval-2026: Few-Shot Arabic LLMs and Complementary-Model Fusion for Stance Detection across Seen and Unseen Targets

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Abstract

We describe our entry to both tracks of StanceEval-2026, an Arabic stance-detection task built on the Mawqif-XT corpus. Track 1 (Seen Targets) is evaluated on Women Driving, while Track 2 (Unseen Targets) is evaluated on Electric Cars and the Trimester school calendar system. Rather than committing to a single approach, we compare two paradigms: fine-tuned Arabic encoders trained with multi-task learning using Mawqif-XT’s sentiment, sarcasm, and rationale annotations, and native Arabic LLMs used in a few-shot setting without any fine-tuning. For each track, we build the final system around the stronger approach. Across more than forty experimental configurations, few-shot LLMs consistently outperform fine-tuned encoders on both tracks. Our final systems achieve an Favg2 of 0.834 on Track 2 (ranked 9th) and 0.747 on Track 1 (ranked 13th), the latter through a complementary fusion of ALLaM-7B and Fanar-9B. We further observe that model performance may be related to how well pretraining coverage aligns with a target’s cultural and linguistic domain, that fusing two comparably strong models can outperform either model individually, and, perhaps most importantly, that more elaborate methods never surpass a strong LLM with a simple prompt. We release our code and prompts to support reproducibility.

1 Introduction

Stance detection asks whether a text is in favour of, against, or neutral toward a given target (Mohammad et al., 2016). It is often harder than sentiment analysis because the target is not always stated explicitly. For Arabic, the problem remains under-resourced, and models tend to be less robust on targets they did not see during training (Cruickshank and Ng, 2026).

We participated in both tracks of StanceEval-2026 (Albalawi et al., 2026a), which uses the

Mawqif-XT corpus (Albalawi et al., 2026b). Track 1 (Seen Targets) is evaluated on Women Driving, a target from a domain already represented in the training data, while Track 2 (Unseen Targets) is evaluated on Electric Cars and the Trimester school calendar system. Rather than committing to a single model, we conduct a controlled comparison between two approaches: fine-tuned Arabic encoders trained with multi-task learning that leverages Mawqif-XT’s sentiment, sarcasm, and a novel stance-explicitness signal, and native Arabic LLMs used in a few-shot setting with a fixed set of three examples and no fine-tuning. For each track, we build the final system around the stronger approach and apply model fusion where beneficial. We report everything we tried, including what failed.

Our contributions are: (i) a direct comparison showing that few-shot Arabic LLMs consistently outperform fine-tuned encoders on both tracks; (ii) an analysis suggesting that a model’s advantage may depend on how well its pretraining covers the target’s cultural and linguistic domain; (iii) a fusion of two comparably strong LLMs that outperforms either model individually; and (iv) a broad catalogue of negative results spanning fine-tuning, distillation, synthetic augmentation, target masking, reasoning-based prompting, and multi-agent approaches, none of which surpasses a strong few-shot LLM with a simple prompt. Our final systems achieve an Favg2 of 0.834 on Track 2 (ranked 9th) and 0.747 on Track 1 (ranked 13th). On Track 2, our best system surpasses all three official baselines (MARBERT 0.722, JAIS-70B 0.691, and the eight-times-larger Qwen2.5-72B 0.748). We release our code and prompts at <https://github.com/Saliyah-53/saliyah-stanceeval2026>.

2 Related Work

Prior Arabic stance detection has been dominated by fine-tuned encoders. Pretrained Arabic trans-

formers such as AraBERT (Antoun et al., 2020) and the dialect-oriented MARBERT (Abdul-Mageed et al., 2021) provide strong baselines for social-media text, and the Mawqif dataset (Alturayef et al., 2022), which jointly annotates stance, sentiment, and sarcasm, enabled multi-task stance models (Alturayef et al., 2023). This line of work culminated in StanceEval-2024, whose top-ranked system combined multi-task learning, contrastive losses, and ensembling (Badran et al., 2024). StanceEval-2026 extends this to unseen, cross-target evaluation (Albalawi et al., 2026a).

In parallel, large language models (LLMs) have been applied to cross-target and zero-shot stance detection through prompting and external-knowledge augmentation (Fan et al., 2025; Cruickshank and Ng, 2026). Several capable native Arabic LLMs have since emerged, including Jais (Sengupta et al., 2023), ALLaM (Bari et al., 2025), Fanar (Team et al., 2025), and SILMA (SILMA AI, 2024). Their comparative performance on Arabic stance detection, however, remains largely unexplored. We address this gap with a controlled comparison across both tracks, showing that few-shot native Arabic LLMs consistently outperform fine-tuned encoders and that fusing two complementary LLMs outperforms either individually.

3 Data and Metrics

We use Mawqif-XT, the corpus released for StanceEval-2026 (Albalawi et al., 2026b,a) as an extension of Mawqif (Alturayef et al., 2022), in which each tweet is annotated for stance (Favor, Against, or None) together with sentiment, sarcasm, annotator confidence, and stance-rationale fields. The training and development data cover three seen targets (COVID-19 Vaccine, Digital Transformation, and Women Empowerment) while the blind test comprises three newly introduced targets (Albalawi et al., 2026b). Under the two-track design (Albalawi et al., 2026a), Track 1 (Seen Targets) is evaluated on Women Driving (352 tweets), a target thematically adjacent to the seen Women Empowerment, whereas Track 2 (Unseen Targets) is evaluated on Electric Cars (332 tweets) and Trimester (312 tweets), totaling 644, which are unrelated to any training target. The primary metric is Favg2, the macro-averaged F_1 over the Favor and Against classes computed per target and then averaged; the secondary metric Favg3 additionally includes None (Albalawi et al., 2026a). All blind-test scores are

official Codabench submissions; ablations and the description study use the development set. We never accessed the gold test labels.

4 Systems

Multi-task encoder. We encode each target and tweet with a shared Arabic transformer that has four output heads: a main stance head plus three auxiliary ones for sentiment, sarcasm, and a new binary *stance-explicitness* signal we take from the rationale fields. The loss adds a confidence-weighted stance term to masked auxiliary terms (Appendix B). We fine-tune four encoders (MARBERTv2, AraBERT-large, AraELECTRA, and CAMeLBERT-sentiment), and we compare a simple ad-hoc cleaner with the official AraBERT pre-processor.

Few-shot LLMs. We test four native Arabic LLMs (Fanar-9B, SILMA-9B, Jais-13B, and ALLaM-7B; exact checkpoints are listed in Appendix A), all loaded in 4-bit and run without any fine-tuning. The prompt is a fixed 3-shot setup: a unified Arabic instruction, three fixed labelled examples (one per stance class, taken from a training target), and an optional one-line description of the test target, which helps most on lexically ambiguous targets (Appendix G). Each model returns a single stance word, which a small parser maps to the label set. On top of this base prompt we try three inference-time strategies. Chain-of-thought asks the model to reason step by step before it commits to a label. Self-consistency samples several such chains and takes the majority vote. A *guidance-rich* prompt adds short warnings about the pitfalls we kept seeing (dialectal spelling, sarcasm, quoting an opposing view, and hashtags that mark the topic rather than the author’s stance), while still asking for a direct one-word answer. All prompts are in Appendix D.

Fusion. Our final Track 1 system combines two LLM prediction files by voting. When the two agree, we take their shared label. When they disagree, we go with the higher performing model and avoid None, which tended to be over-predicted on those disagreements (Appendix E).

5 Experimental Setup

For the encoder models, we use a maximum sequence length of 128, AdamW optimization with a learning rate of 2×10^{-5} , and 10 training epochs,

selecting the best checkpoint by development-set Favg2 (seed 42); each encoder is trained on the official training split of its track. The LLMs are loaded in 4-bit using HuggingFace Transformers and bitsandbytes, decoded greedily ($T=0$) for direct prediction and with temperature sampling ($T \in [0.3, 0.9]$, five votes) for self-consistency. Text is preprocessed with the official AraBERT preprocessor. All experiments are performed on a single NVIDIA T4 GPU. Results are reported from a single run (seed 42); differences below approximately 0.005 in Favg2 should be regarded as indicative rather than statistically significant. Full hyper-parameters and Arabic-specific preprocessing details are in Appendix C.

6 Results and Analysis

Tables 1 and 2 give the headline results per track; full tables and ablations are in Appendix F. Several consistent patterns emerge from the experiments.

Track 2 (Unseen) system	Favg2
MARBERT (official baseline)	0.722
JAIS-70B (official, zero-shot)	0.691
Qwen2.5-72B (official, zero-shot)	0.748
AraBERT-large + preproc. (best encoder)	0.694
SILMA-9B (few-shot)	0.791
Fanar-9B (few-shot)	0.828
Fanar-9B (guidance prompt)	0.834

Table 1: Track 2 headline results (Favg2). Full table: Appendix F.

Track 1 (Seen) system	Favg2
MARBERTv2 + preproc. (best trained)	0.685
Fanar-9B (few-shot)	0.715
ALLaM-7B (CoT + self-consistency)	0.726
Triple fusion (ALLaM+Fanar+SILMA)	0.717
Fusion ALLaM+Fanar (simple prompt)	0.743
Fusion ALLaM+Fanar (guidance)	0.747

Table 2: Track 1 headline results (Favg2). Full table: Appendix F.

Few-shot LLMs outperform fine-tuned encoders.

Encoders peak at 0.694 (Track 2) and 0.685 (Track 1), while the best few-shot LLM reaches 0.834 and 0.726. The gain comes from broad pre-trained knowledge rather than scale, since a 9B model beats a 72B one (the official baselines are evaluated zero-shot, whereas our systems use a fixed 3-shot prompt). It also disappears under narrow fine-tuning: ALLaM QLoRA drops to 0.412 on Track 2.

Pretraining and target alignment. The leading LLM flips between tracks. On the general Track 2, Fanar beats ALLaM (0.828 vs 0.682); on the Saudi-rooted Women Driving target, ALLaM (developed by SDAIA with strong Arabic dialectal and cultural coverage) beats Fanar by a small margin (0.726 vs 0.715). We read this as an alignment between pretraining coverage and the target’s domain, not as an effect of model origin. Because this margin is small and rests on a single target, we treat it only as a hypothesis for future work.

Complementary fusion. On Track 1, ALLaM (0.726) and Fanar (0.715) agree on only 70.5% of tweets. Fusing them reaches 0.743, while adding the weaker SILMA (triple fusion) drops to 0.717. A guidance-rich prompt lowered the individual models yet raised their fusion to 0.747 by making their errors more complementary. The 0.747 vs 0.743 gap is within single-run noise; the robust effect is the fusion’s gain of roughly two points over its stronger component (ALLaM, 0.726).

Simplicity wins; complexity hurts. Bare chain-of-thought reduced accuracy (ALLaM 0.723→0.687); self-consistency helped a weak model only within noise (0.723→0.726) while clearly hurting a confident one (Fanar 0.715→0.691). Three elaborate prompting schemes (mixture-of-experts, sarcasm staging, and multi-agent debate) all underperformed the simple prompt across both models (debate worst: Fanar 0.647, ALLaM 0.469; Appendix H). Surface rules also mislead: an explicit-hashtag override collapsed Track 1 to 0.658, since a hashtag marks the topic of debate, not the author’s stance.

Why the unseen track is easier. Our best Track 2 (0.834) exceeds Track 1 (0.747) despite the target being new, because Women Driving discourse is far more complex: 100% hashtags, 46.6% emojis, and 16.2% sarcasm cues versus 5.1% in Track 2 (Appendix I). This suggests that discourse complexity may be a stronger source of difficulty than target novelty.

Transferring LLM gains to trained models.

None of our attempts succeeded: adding the dev target lowered MARBERT (0.685→0.667), distillation reached 0.615, and synthetic augmentation 0.649, all below the plain trained model.

7 Conclusion

We described our submission to both tracks of StanceEval-2026, which we set up as a comparison between fine-tuned Arabic encoders and few-shot native Arabic LLMs. The few-shot LLMs won on both tracks. Our best systems reach Favg2 0.834 on Track 2 (rank 9) and 0.747 on Track 1 (rank 13, using an ALLaM and Fanar fusion). We also tried many things that did not help: neither narrow fine-tuning nor more elaborate inference-time methods ever beat a strong few-shot LLM with a simple prompt. We release our code and prompts, and we leave the link between a model’s pretraining coverage and its per-target performance as an open question for future work.

Limitations

All of our results come from a single run (seed 42); we did not average over several seeds, so small gaps (below about 0.005 Favg2) are only indicative, not statistically significant, and our fusion and self-consistency numbers reuse these single-run predictions. The pretraining and target alignment observation rests on just one culturally specific target, so we treat it as a hypothesis rather than an established result. All of our models are 7 to 13B in 4-bit on a single T4 GPU; larger models or full-precision inference could change the picture. Finally, the study stays within Arabic and the Mawqif-XT tweet domain, and our discourse analysis relies on shallow surface features that approximate, but do not fully capture, how hard a tweet is to interpret.

Ethics Statement

We use the publicly released Mawqif-XT corpus under the shared-task terms. We collected no new data and never accessed the blind-test gold labels. Stance detection on socially sensitive targets can be misused for opinion surveillance, so we report only aggregate results and keep all findings at the corpus level. The LLMs may carry social and cultural biases, and our observation about pretraining coverage concerns the training data, not a model’s origin, and takes no political position. We follow the ACL Ethics Policy.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. [ARBERT &](#)

[MARBERT: Deep bidirectional transformers for Arabic](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Rasha Albalawi, Nuha Albadi, Hamzah Luqman, Saad Ezzini, Maram Kurdi, Asma Yamani, Ahmed Ashraf, Maged Al-Shaibani, and Nora Alturayef. 2026a. Stanceeval-2026: The second stance detection shared task. In *Proceedings of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Rasha Albalawi, Nuha Albadi, Hamzah Luqman, Maram Kurdi, Saad Ezzini, Asma Yamani, and Ahmed Ashraf. 2026b. [Mawqif-xt: An arabic benchmark dataset for cross-target stance detection](#). Preprint, arXiv:2608.09539.

Nora Alturayef, Hamzah Luqman, and Moataz Ahmed. 2023. [Enhancing stance detection through sequential weighted multi-task learning](#). *Social Network Analysis and Mining*, 14(1):7.

Nora Saleh Alturayef, Hamzah Abdullah Luqman, and Moataz Aly Kamaleldin Ahmed. 2022. [Mawqif: A multi-label Arabic dataset for target-specific stance detection](#). In *Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP)*, pages 174–184, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. [AraBERT: Transformer-based model for Arabic language understanding](#). In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resources Association.

Mohamed Badran, Mo’men Hamdy, Marwan Torki, and Nagwa El-Makky. 2024. [AlexUNLP-BH at StanceEval2024: Multiple contrastive losses ensemble strategy with multi-task learning for stance detection in Arabic](#). In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 823–827, Bangkok, Thailand. Association for Computational Linguistics.

M Saiful Bari, Yazeed Alnumay, Norah A. Alzahrani, Nouf M. Alotaibi, Hisham Abdullah Alyahya, Sultan AlRashed, Faisal Abdulrahman Mirza, Shaykhah Z. Alsubaie, Hassan A. Alahmed, Ghadah Alabduljabbar, Raghad Alkhathran, Yousef Almushayqih, Raneem Alnajim, Salman Alsubaihi, Maryam Al Mansour, Saad Amin Hassan, Dr. Majed Alrubaian, Ali Alammari, Zaki Alawami, and 7 others. 2025. [AL-Lam: Large language models for arabic and english](#). In *The Thirteenth International Conference on Learning Representations*.

Iain J. Cruickshank and Lynnette Hui Xian Ng. 2026. [Prompting and fine-tuning open source large language models for stance classification](#). *ACM Trans. Intell. Syst. Technol.*, 17(4).

Qinlong Fan, Jicang Lu, Yepeng Sun, Qiankun Pi, and Shouxin Shang. 2025. [Enhancing zero-shot stance detection via multi-task fine-tuning with debate data and knowledge augmentation](#). *Complex & Intelligent Systems*, 11:151.

Saif Mohammad, Svetlana Kiritchenko, Parinaz Sobhani, Xiaodan Zhu, and Colin Cherry. 2016. [SemEval-2016 task 6: Detecting stance in tweets](#). In *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, pages 31–41, San Diego, California. Association for Computational Linguistics.

Neha Sengupta, Sunil Kumar Sahu, Bokang Jia, Satheesh Katipomu, Haonan Li, Fajri Koto, William Marshall, Gurpreet Gosal, Cynthia Liu, Zhiming Chen, Osama Mohammed Afzal, Samta Kamboj, Onkar Pandit, Rahul Pal, Lalit Pradhan, Zain Muhammad Mujahid, Massa Baali, Xudong Han, Soudos Mahmoud Bsharat, and 13 others. 2023. [Jais and jais-chat: Arabic-centric foundation and instruction-tuned open generative large language models](#).

SILMA AI. 2024. [SILMA-9B-Instruct-v1.0](#). Hugging Face model card.

Fanar Team, Ummar Abbas, Mohammad Shahmeer Ahmad, Firoj Alam, Enes Altinisik, Ehsannedin Asgari, Yazan Boshmaf, Sabri Boughorbel, Sanjay Chawla, Shammur Chowdhury, Fahim Dalvi, Kareem Darwish, Nadir Durrani, Mohamed Elfeky, Ahmed Elmagarmid, Mohamed Eltabakh, Masoomali Fatehkia, Anastasios Fragkopoulos, Maram Hasanain, and 23 others. 2025. [Fanar: An arabic-centric multimodal generative ai platform](#).

A Exact Model Versions

All LLMs are loaded in 4-bit. Hugging Face checkpoints:

- Fanar-9B:
QCRI/Fanar-1-9B-Instruct
- ALLaM-7B:
humain-ai/ALLaM-7B-Instruct-preview
- SILMA-9B:
silma-ai/SILMA-9B-Instruct-v1.0
- Jais-13B:
inceptionai/jais-13b-chat

Encoders: MARBERTv2, AraBERT-large, Ara-ELECTRA, and CAMELBERT-sentiment.

B Multi-Task Objective

Following the weighted multi-task learning of [Alturayef et al. \(2023\)](#), and extending it with annotator-confidence instance weighting and a stance-explicitness head, our training objective is

$$\mathcal{L} = \frac{1}{N} \sum_i w_i \text{CE}(g_{\text{st}}(h_i), y_i) + \sum_k \lambda_k \text{CE}_{\text{mask}}(g_k(h_i), y_i^k),$$

with $w_i = 0.5 + 0.5c_i$ the annotator-confidence weight and $\lambda_k \in \{0.3, 0.2, 0.2\}$ for sentiment, sarcasm, and explicitness, which were set manually to give the primary stance task the largest weight and the auxiliary tasks smaller supporting weights. A stance is labelled *explicit* when its rationale field names the stance target directly and *implicit* otherwise, giving 2,853 explicit versus 257 implicit training instances.

C Setup and Arabic-Specific Details

Encoders: batch size 16–32 (AraBERT-large 8), 10% warmup, dropout 0.2, gradient clipping 1.0. Preprocessing uses the official AraBERT preprocessor: normalization of alef/ya and tamarbuta, tatweel and diacritic removal, and mention/URL/hashtag/emoji handling; tokenization uses each model’s own tokenizer. LLM prompts are written in Arabic; self-consistency uses five samples with temperatures $\{0.3, 0.5, 0.7, 0.9, 0.6\}$. We also tested test-time MSA rewriting of dialectal input, which changed 0/352 Fanar predictions on Track 1, indicating dialect-invariance on this task.

D Prompts

Prompts were issued in Arabic; English glosses follow (exact strings are released with our code). All few-shot runs share **three fixed labelled demonstrations**, selected once from the training split (one per class, from the seen COVID-19 Vaccine target) and reused unchanged across all experiments: (1) “*Vaccination saved millions of lives and everyone must take it*” → Favor; (2) “*I don’t trust the vaccine, it has many harms, a conspiracy*” → Against; (3) “*When do vaccination centres open?*” → None.

Unified (base) instruction: “You are an expert analyst of Arabic social-media text. Decide the author’s stance toward the target and answer with one word: Favor, Against, or None.”

Guidance-rich instruction: additionally reminds the model to account for dialect (Gulf/Hijazi/Najdi), colloquialisms, sarcasm and irony; to distinguish the author’s own view from a quoted or mocked one (the stance may oppose the surface words or hashtag); and that a hashtag marks the topic of debate, not necessarily the author’s stance. It then requires a single English word with no explanation.

E Fusion Rule

For a stronger model M_s (ALLaM on Track 1) and a weaker M_w : if $M_s=M_w$ take it; if exactly one predicts None take the other; on any remaining disagreement take M_s . The two agree on 70.5% of the 352 Track 1 tweets.

F Full Result Tables

Table 3: Full Track 2 blind-test results (Favg2).

Track 2 system	Favg2
MARBERT (baseline)	0.722
JAIS-70B (baseline, zero-shot)	0.691
Qwen2.5-72B (baseline, zero-shot)	0.748
MARBERTv2 (ad-hoc preproc.)	0.416
AraELECTRA + preproc.	0.631
CAMeLBERT-sent. + preproc.	0.643
MARBERTv2 + preproc.	0.672
AraBERT-large + preproc.	0.694
ALLaM-7B (QLoRA)	0.412
ALLaM-7B (few-shot)	0.682
ALLaM-7B (CoT+SC)	0.701
Jais-13B (few-shot)	0.701
SILMA-9B (few-shot)	0.791
Fanar-9B (few-shot)	0.828
Fanar-9B (guidance)	0.834

Table 4: Full Track 1 blind-test results (Favg2).

Track 1 system	Favg2
MARBERTv2 + preproc. (train only)	0.685
MARBERTv2 (train + dev)	0.667
AraBERT-large + preproc.	0.661
Synthetic augmentation (Fanar)	0.649
Transductive distillation (Fanar)	0.615
SILMA-9B (few-shot)	0.698
Fanar-9B (few-shot / guidance)	0.715
ALLaM-7B (direct)	0.723
ALLaM-7B (bare CoT)	0.687
ALLaM-7B (CoT+SC, 5)	0.726
ALLaM-7B (SC, 9)	0.721
Hashtag-override rule	0.658
Test-time MSA rewriting	0.696
Triple fusion (+SILMA)	0.717
Fusion ALLaM+Fanar (simple)	0.743
Fusion ALLaM+Fanar (guidance)	0.747

Table 5: Leave-one-out ablation (dev, Favg2 $\times$ 100).

Ablation (dev)	Favg2	Δ
Full MTL system	81.9	—
– explicitness head	81.2	−0.7
– sentiment head	80.9	−1.0
– sarcasm head	80.5	−1.4

G Target-Description Effect

Table 6: Effect of target-description injection (Track 2, Favg2).

LLM	with desc.	no desc.	Δ
Fanar-9B	0.828	0.670	+0.158
SILMA-9B	0.791	0.729	+0.062
Jais-13B	0.701	0.562	+0.139
ALLaM-7B	0.682	0.594	+0.088

H Complex Prompting

Table 7: Complex prompting vs. the simple prompt (Track 2, Favg2).

Strategy (Track 2)	Fanar	ALLaM
Guidance prompt (simple)	0.834	0.714
Mixture-of-experts	0.740	0.691
Sarcasm-aware two-stage	0.721	0.665
Multi-agent debate	0.647	0.469

I Discourse Features

Table 8: Surface discourse features of the two blind-test sets.

Feature	Track 1	Track 2
Hashtags	100.0%	14.9%
Emojis	46.6%	18.3%
Sarcasm / laughter cues	16.2%	5.1%
Mentions (replies)	0.9%	65.2%
Avg. length (chars)	100	139